

1. Scenario Review

Project: Telemedicine Slot Booking & Prescription Portal

Core functionality:

- Patients can browse physicians by specialty and view available slots.
- Patients can book video consultations.
- The system generates secure consultation links.
- Physicians can author and digitally sign prescriptions.
- Patients can download prescriptions as PDFs.
- Both patients and physicians receive booking notifications.

Key architectural challenges:

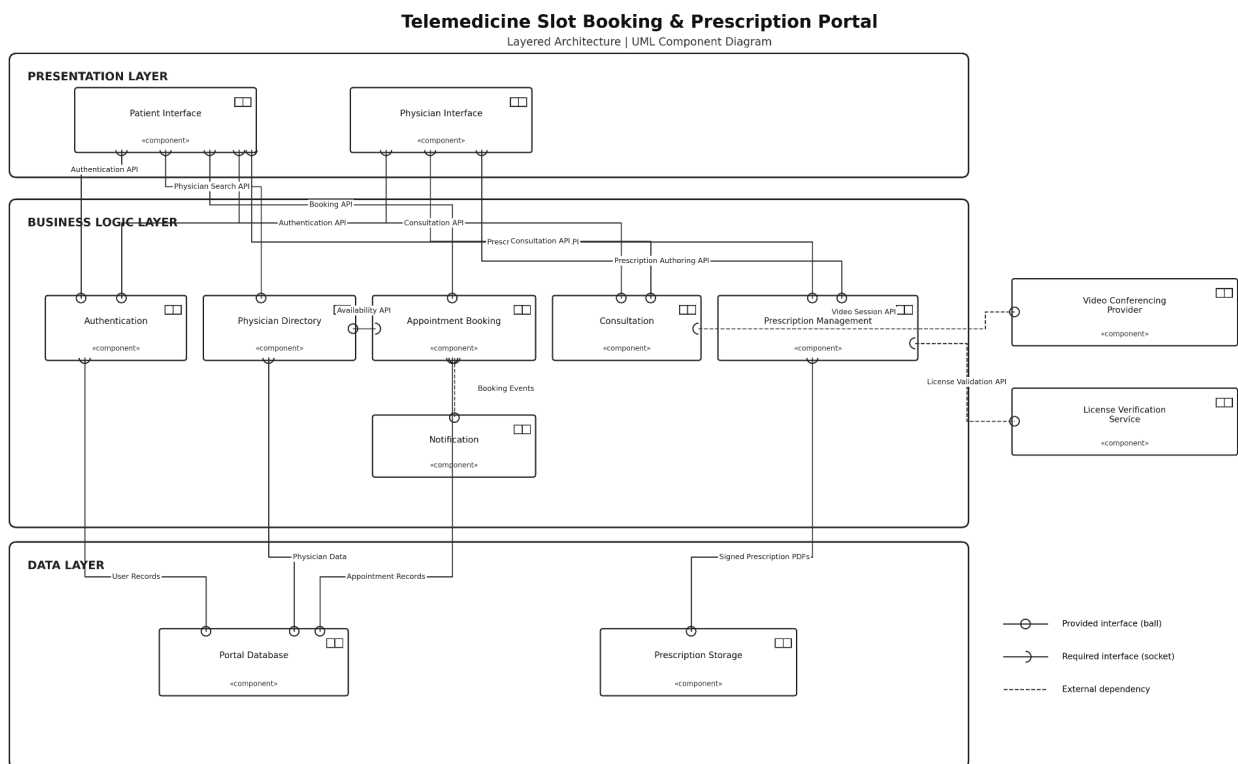
1. **Security:** Protect patient data, clinical notes and consultation links.
2. **Performance:** Ensure quick physician searches and appointment bookings.
3. **Availability:** Maintain reliable access to booking and consultation services.
4. **Maintainability:** Keep booking, consultation and prescription functionality organized.

2. Architectural Style Analysis

Architecture	Advantages	Disadvantages
Layered	Clear separation of concerns, easy to understand and maintain.	Performance overhead and difficulty scaling individual components.
Microservices	Independent scaling, fault isolation and independent deployment.	Operational complexity, network latency and data consistency challenges.
Client-Server	Centralized control, simple deployment and easier data consistency.	Single point of failure, scalability bottlenecks and limited fault tolerance.

The Telemedicine Slot Booking & Prescription Portal requires secure handling of patient data, reliable appointment booking and efficient management of consultations and prescriptions. The three architectural styles offer different approaches to meeting these requirements. Layered Architecture separates the system into distinct layers, Microservices Architecture allows independent deployment and scaling, while Client-Server Architecture centralizes application processing.

3. Architecture Selection & Component Identification



Selected Architecture: Layered Architecture

The Telemedicine Slot Booking & Prescription Portal will use Layered Architecture, consisting of Presentation, Business Logic and Data layers.

Identified Components

1. Presentation Layer

- Patient Interface: Allows patients to browse physicians, book consultations, join video sessions and download prescriptions.
- Physician Interface: Allows physicians to manage consultation slots, conduct consultations and create digitally signed prescriptions.

2. Business Logic Layer

- Authentication Component: Handles user authentication and access control.
- Physician Directory Component: Manages physician profiles, specialties and available slots.
- Appointment Booking Component: Handles slot selection, booking confirmation and appointment management.
- Consultation Component: Manages secure consultation sessions and access links.
- Prescription Management Component: Handles prescription creation, digital signing and PDF generation.
- Notification Component: Sends booking confirmations and updates to patients and physicians.

3. Data Layer

- Database Component: Stores user information, physician profiles and appointment records.
- Prescription Storage Component: Stores digitally signed prescriptions securely.

4. Architectural Pattern Selection

Selected Architectural Pattern: Layered Architecture (Three-Tier Architecture)

The Telemedicine Slot Booking & Prescription Portal uses a three-tier layered architecture.

- **Presentation Layer:** Provides interfaces for patients and physicians to interact with the system.
- **Business Logic Layer:** Handles authentication, physician search, appointment booking, consultations, prescription management and notifications.
- **Data Layer:** Manages user information, physician profiles, appointment records and prescription storage.

The three layers separate user interaction, business operations and data management, with each layer having a specific responsibility.

5. Architectural Justification

1. Scenario-Specific Reason 1: Security and Privacy

The telemedicine portal handles sensitive patient information, clinical notes and digitally signed prescriptions. Layered Architecture allows authentication and access control to be handled by dedicated components, while sensitive data is managed separately in the data layer. This supports secure handling of patient information.

2. Scenario-Specific Reason 2: Modularity

The system includes several independent functions, such as physician search, appointment booking, video consultations and prescription management. Layered Architecture separates these responsibilities, making individual components easier to modify and maintain without affecting the entire system.

3. Security Advantage

The separation of presentation, business logic and data layers allows security controls to be applied at different levels. Authentication, access control and encryption can be implemented to protect patient records and prescription data from unauthorized access.

4. Performance Benefit

Separating business logic from the presentation layer allows frequently used operations, such as physician searches and appointment bookings, to be optimized independently. This can help reduce response times and improve the overall responsiveness of the portal.